

Threat-Intelligence Kill-Chain Correlation Engine

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

in

A.Y. 2026-2027

Submitted by

V. Hasith Varma	2520090125
J. Amrutha Varshini	2520080066

Team Number: 8

Section: 12

Under the guidance of

Ms.K.Chandusha

Asst. Prof., CS&IT



Hyderabad-500090, Telangana, India.

ABSTRACT

The Threat-Intelligence Kill-Chain Correlation Engine is developed to help Security Operations Centres (SOCs) analyse large amounts of security logs and alerts more efficiently. Instead of manually checking thousands of log entries, the system automatically identifies known attack patterns and connects related events to recreate the complete sequence of a cyberattack, known as a kill chain.

It represents attack sequences as graph structures to identify critical choke points where defensive actions can effectively interrupt the attack. In addition, the system recommends the minimum number of monitoring points required to cover all possible attack paths and assigns detected alerts to security analysts based on priority. This helps security teams understand the progression of an attack more clearly and focus their attention on the most important security events.

The system is designed to reduce the manual effort involved in analysing security data and make the overall threat analysis process more organized and efficient. By connecting individual alerts into meaningful attack sequences, it provides a clearer view of how an attack develops across different systems.

By applying Data Structures and Algorithms such as hashing, graph traversal, pattern matching, and graph optimization techniques, this project demonstrates how DSA concepts can solve real-world cybersecurity challenges while improving the speed and accuracy of threat detection and analysis.

Signature of the Guide